ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 6

NATURAL SCIENCES AND TECHNOLOGY

PAPER 1 · PRACTICE B · MARKING GUIDELINE

MARKS: 70

Marking guideline for the Academic Central Grade 6 Natural Sciences and Technology paper 1 · practice b.

Written by Academic Central for practice, following the CAPS curriculum for Grade 6. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept answers in the learner's own words if the science is correct. Do not take marks off for spelling if the meaning is clear.

QUESTION 1

1 1.1 B ✓ 1.2 B ✓ 1.3 B ✓ 1.4 B ✓ 1.5 B ✓ 1.6 B ✓ 1.7 C ✓ 1.8 A ✓ 1.9 A ✓ 1.10 A ✓ (10)

[10]

QUESTION 2

2 2.1 TRUE ✓ 2.2 FALSE ✓ 2.3 TRUE ✓ 2.4 FALSE ✓ 2.5 TRUE ✓ 2.6 TRUE ✓ 2.7 FALSE ✓ 2.8 TRUE ✓ (8)

[8]

QUESTION 3

3.1 Cell ✓; switch ✓; bulb (light bulb) ✓ (3)

3.2 Circuit Q ✓. The switch is open, so the circuit is not complete and electricity cannot flow ✓ (2)

3.3 Less brightly ✓. The energy from one cell is shared between two bulbs ✓ (2)

3.4 Another cell (a second cell) ✓ (1)

3.5 It is the energy source: it pushes electricity around the circuit ✓ (1)

3.6 Electricity must flow in a complete path ✓ from one end (terminal) of the cell, through the bulb and back to the other end ✓ (2)

3.7.1 Copper wire ✓; aluminium foil ✓; steel paper clip ✓ (3)

3.7.2 Metal ✓ (1)

3.7.3 Plastic and rubber are insulators, so they protect the electrician from an electric shock ✓ (1)

3.8 Light ✓ and heat ✓ (2)

[18]

QUESTION 4

4.1 4.1.1 heat ✓ 4.1.2 movement (kinetic) ✓ 4.1.3 electrical ✓ (3)

4.2 Eskom ✓ (1)

4.3 Advantage: the Sun's energy is free / renewable / does not pollute ✓. Disadvantage: panels cost a lot to buy / they make little electricity at night or on cloudy days ✓ (2)

4.4 Too many plugs are in one socket, so the multi-plug is overloaded ✓. It can get too hot and start a fire ✓ (2)

4.5 Any TWO: never fly kites near power lines ✓; never climb pylons or trees near power lines ✓; stay away from fallen wires and report them; do not touch anything that touches a power line (2)

4.6 Any TWO: do not touch the person ✓; switch off the power at the plug or the main switch ✓; push the person away with something dry that does not conduct, such as a wooden broom; call for help (an adult or 112) (2)

[12]

QUESTION 5

5.1.1 Mercury ✓ (1)

5.1.2 The further a planet is from the Sun ✓, the longer it takes to move around the Sun ✓ (2)

5.1.3 About 12 years ✓ (1)

5.1.4 It is the planet closest to the Sun ✓ (1)

5.2 The Sun does not move across our sky; the Earth spins (rotates) on its axis ✓ from west to east, so the Sun seems to move from east to west ✓ (2)

5.3.1 Rocks from space (meteorites) crashing into the Moon ✓ (1)

5.3.2 It reflects light from the Sun ✓ (1)

5.4 A star makes its own light (and heat); a planet does not, it reflects light / a planet moves around a star ✓ (1)

5.5 Any TWO: city lights make the sky too bright to see faint stars ✓; the air is clearer and thinner high up ✓; there are fewer clouds and less dust and pollution (2)

5.6 Signals from cellphones, radios and other electrical machines would disturb the faint radio signals from space ✓ (1)

5.7 Any TWO: drive over rocks and sand ✓; take photographs ✓; collect and test soil and rocks; send information back to Earth; make electricity from sunlight (2)

5.8 Mars is very much further away from the Earth than the Moon ✓ (1)

5.9 Any TWO: they float because there is no gravity ✓; their muscles and bones get weak ✓; they cannot get fresh food or water easily; they miss their families; there is no fresh air, so all air must be made or brought (2)

5.10 Any TWO: weather forecasts ✓; TV and radio ✓; cellphone and internet signals; GPS maps; photographs of the Earth (2)

5.11 A group of stars that form a pattern (picture) in the night sky ✓. Any ONE: the Southern Cross (Crux); Orion; Scorpius; the Pointers ✓ (2)

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TOTAL: 70

END OF MARKING GUIDELINE